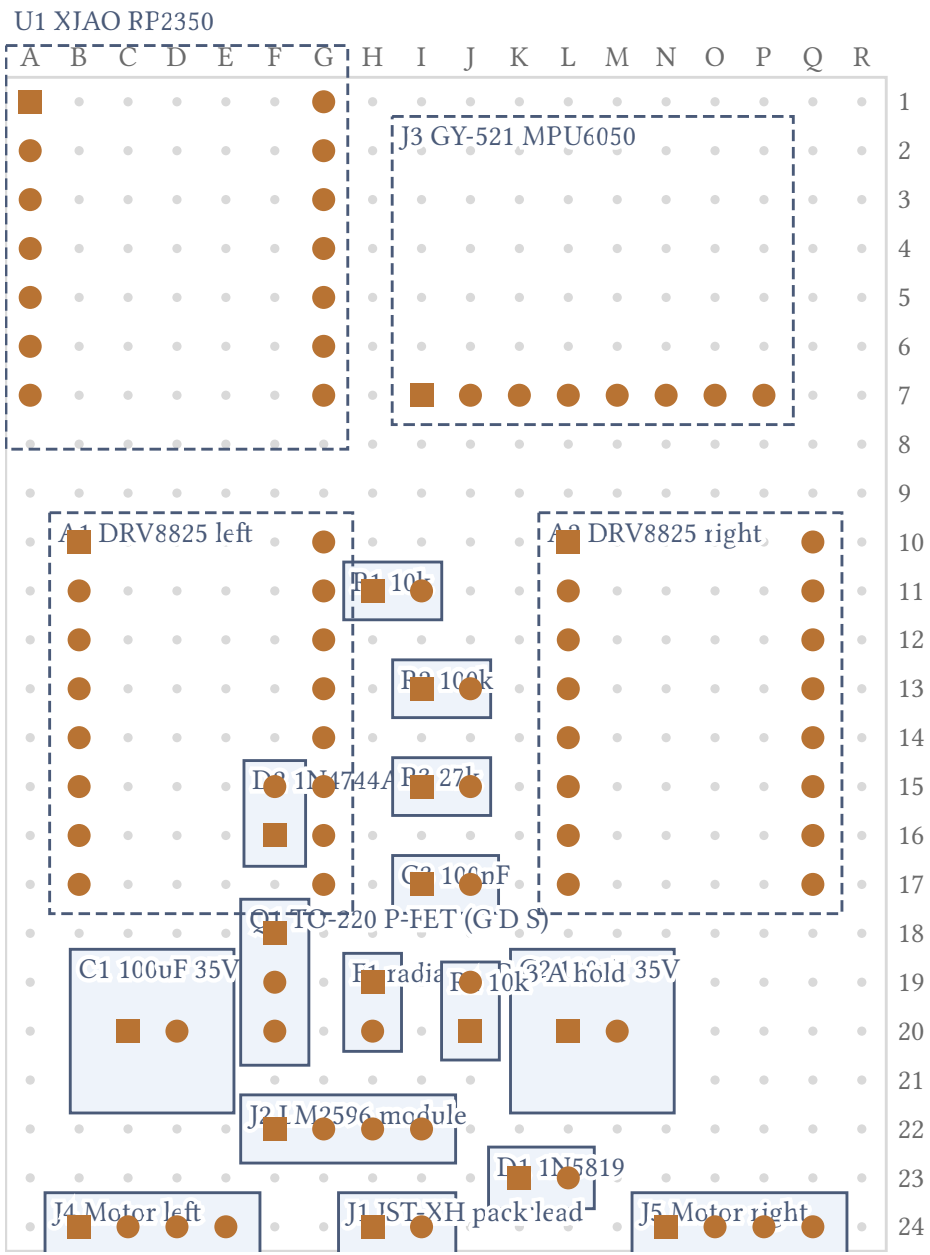
Component side

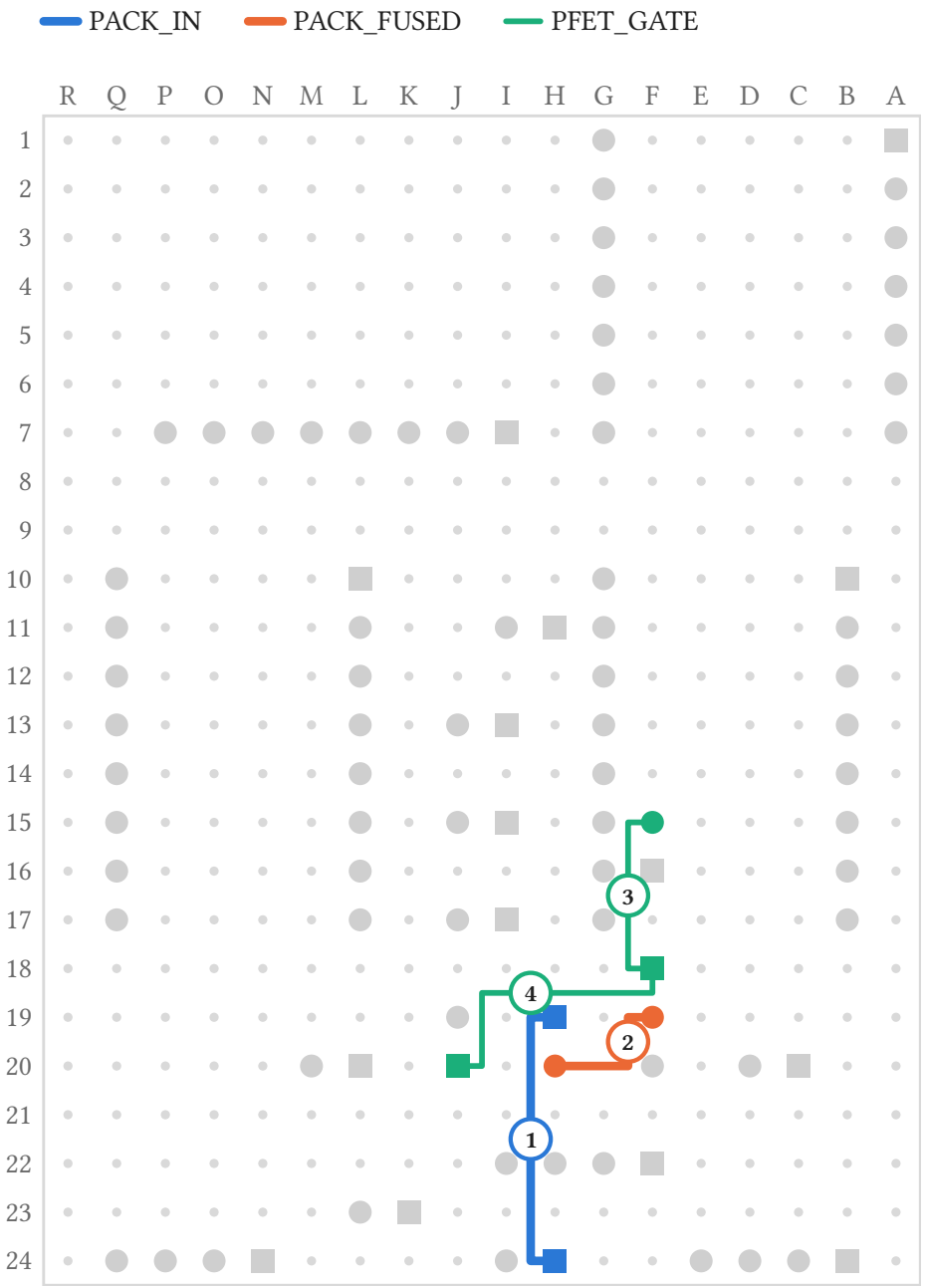


Holes are named from the component side: the solder-side panels are mirrored, the names are not. Each step page lists its wires, numbered as drawn; tick each one off as you solder it, then run that step's continuity checks. Heavy nets (VBAT, GND, motor coils) need 20 AWG or heavier and are drawn thicker; everything else 26–30 AWG.

Parts

Ref	Part	Pin 1 at
A1	DRV8825 left	B10
A2	DRV8825 right	L10
C1	100uF 35V	C20
C2	100uF 35V	L20
C3	100nF	I17
D1	1N5819	K23
D2	1N4744A	F16
F1	radial PTC, 3 A hold	H19
J1	JST-XH pack lead	H24
J2	LM2596 module	F22
J3	GY-521 MPU6050	I7
J4	Motor left	B24
J5	Motor right	N24
Q1	TO-220 P-FET (G D S)	F18
R1	10k	H11
R2	100k	I13
R3	27k	I15
R4	10k	J20
U1	XIAO RP2350	A1

Step 1 · Pack input (wires 1–4)



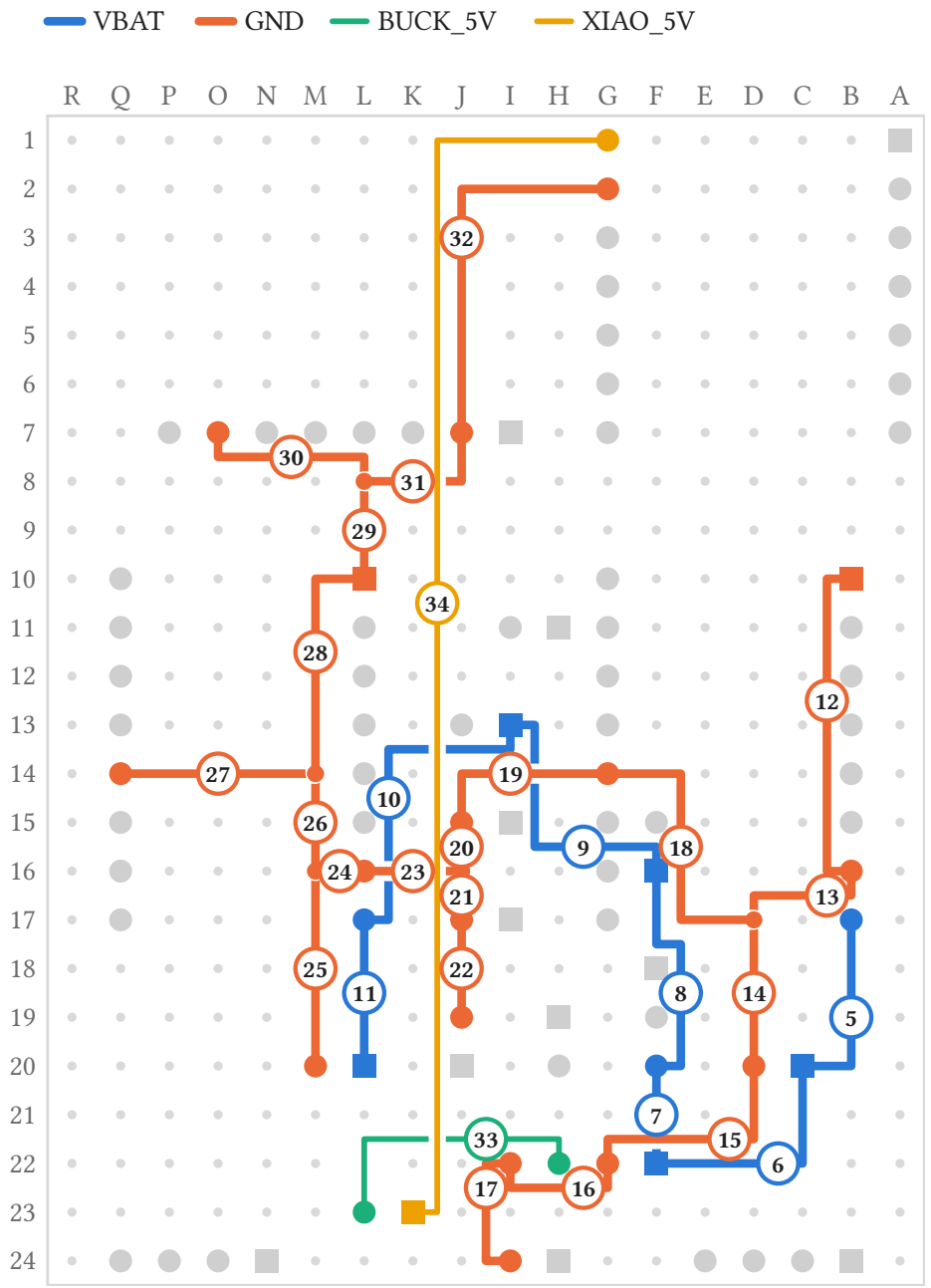
J1 through F1 and the reverse-polarity FET Q1, with its gate zener D2.

	#	Net	Gauge	From	To	Holes
☐	1	PACK_IN	20 AWG	F1.1 (H19)	J1.1 (H24)	6
☐	2	PACK_FUSED	20 AWG	F1.2 (H20)	Q1.2 (F19)	3
☐	3	PFET_GATE	26–30 AWG	D2.2 (F15)	Q1.1 (F18)	4
☐	4	PFET_GATE	26–30 AWG	Q1.1 (F18)	R4.1 (J20)	6

Continuity after this step

	Net	Probe on	Must beep to
☐	PACK_IN	F1.1 (H19)	J1.1 (H24)
☐	PACK_FUSED	F1.2 (H20)	Q1.2 (F19)
☐	PFET_GATE	D2.2 (F15)	Q1.1 (F18), R4.1 (J20)

Step 2 · Power (wires 5–34)



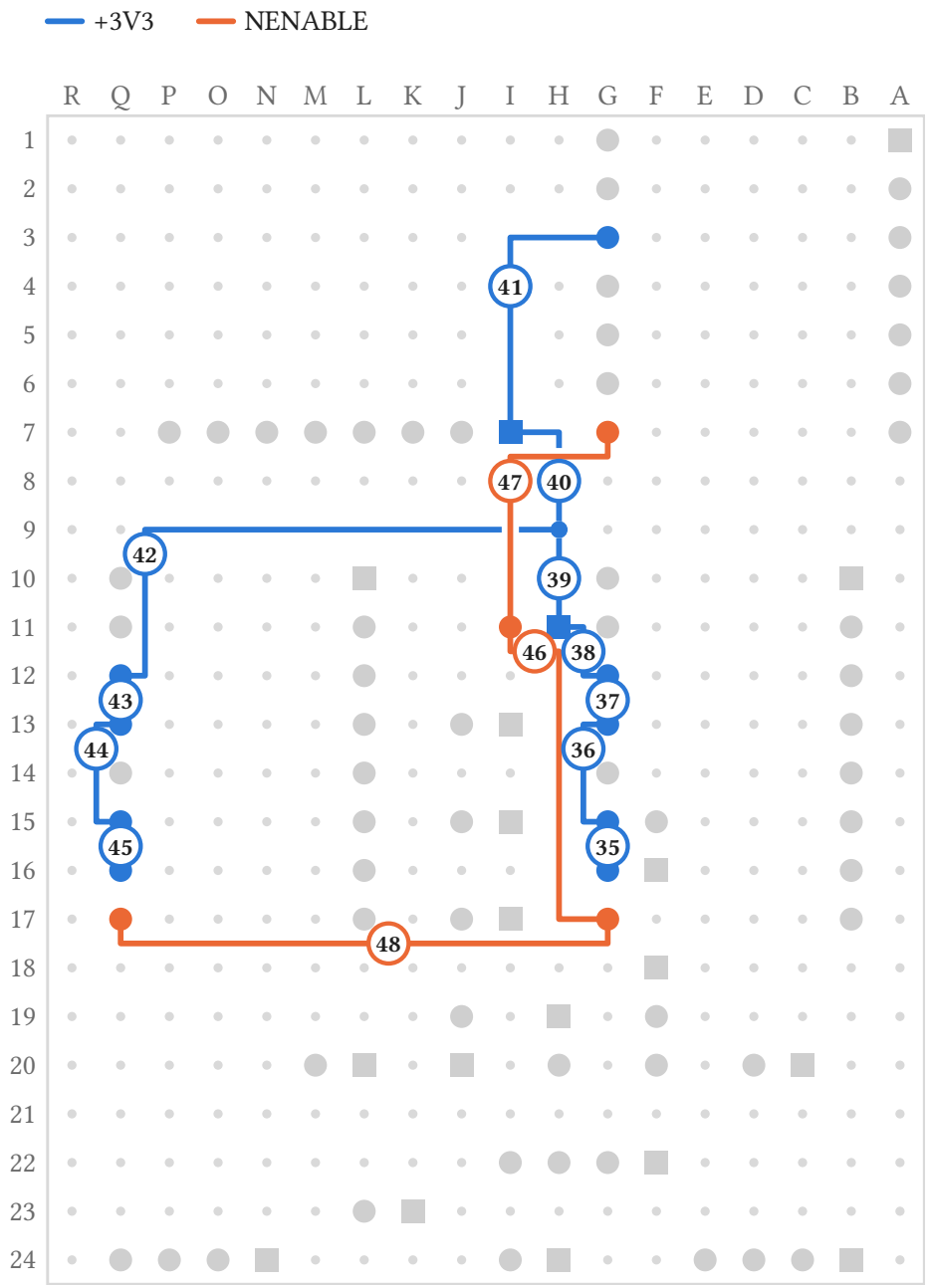
VBAT, GND and the 5 V feed.

	#	Net	Gauge	From	To	Holes
☐	5	VBAT	20 AWG	A1.8 (B17)	C1.1 (C20)	4
☐	6	VBAT	20 AWG	C1.1 (C20)	J2.1 (F22)	5
☐	7	VBAT	20 AWG	J2.1 (F22)	Q1.3 (F20)	2
☐	8	VBAT	20 AWG	Q1.3 (F20)	D2.1 (F16)	5
☐	9	VBAT	20 AWG	D2.1 (F16)	R2.1 (I13)	6
☐	10	VBAT	20 AWG	R2.1 (I13)	A2.8 (L17)	7
☐	11	VBAT	20 AWG	A2.8 (L17)	C2.1 (L20)	3
☐	12	GND	20 AWG	A1.1 (B10)	A1.7 (B16)	7
☐	13	GND	20 AWG	A1.7 (B16)	splice (D17)	3
☐	14	GND	20 AWG	splice (D17)	C1.2 (D20)	3
☐	15	GND	20 AWG	C1.2 (D20)	J2.2 (G22)	5
☐	16	GND	20 AWG	J2.2 (G22)	J2.4 (I22)	3
☐	17	GND	20 AWG	J2.4 (I22)	J1.2 (I24)	3
☐	18	GND	20 AWG	splice (D17)	A1.12 (G14)	6
☐	19	GND	20 AWG	A1.12 (G14)	R3.2 (J15)	4
☐	20	GND	20 AWG	R3.2 (J15)	splice (J16)	1
☐	21	GND	20 AWG	splice (J16)	C3.2 (J17)	1
☐	22	GND	20 AWG	C3.2 (J17)	R4.2 (J19)	2
☐	23	GND	20 AWG	splice (J16)	A2.7 (L16)	2
☐	24	GND	20 AWG	A2.7 (L16)	splice (M16)	1
☐	25	GND	20 AWG	splice (M16)	C2.2 (M20)	4
☐	26	GND	20 AWG	splice (M16)	splice (M14)	2
☐	27	GND	20 AWG	splice (M14)	A2.12 (Q14)	4
☐	28	GND	20 AWG	splice (M14)	A2.1 (L10)	5
☐	29	GND	20 AWG	A2.1 (L10)	splice (L8)	2
☐	30	GND	20 AWG	splice (L8)	J3.7 (O7)	4
☐	31	GND	20 AWG	splice (L8)	J3.2 (J7)	3
☐	32	GND	20 AWG	J3.2 (J7)	U1.13 (G2)	8
☐	33	BUCK_5V	26–30 AWG	D1.2 (L23)	J2.3 (H22)	6
☐	34	XIAO_5V	26–30 AWG	D1.1 (K23)	U1.14 (G1)	26

Continuity after this step

	Net	Probe on	Must beep to
☐	VBAT	A1.8 (B17)	A2.8 (L17), C1.1 (C20), C2.1 (L20), D2.1 (F16), J2.1 (F22), Q1.3 (F20), R2.1 (I13)
☐	GND	A1.1 (B10)	A1.12 (G14), A1.7 (B16), A2.1 (L10), A2.12 (Q14), A2.7 (L16), C1.2 (D20), C2.2 (M20), C3.2 (J17), J1.2 (I24), J2.2 (G22), J2.4 (I22), J3.2 (J7), J3.7 (O7), R3.2 (J15), R4.2 (J19), U1.13 (G2)
☐	BUCK_5V	D1.2 (L23)	J2.3 (H22)
☐	XIAO_5V	D1.1 (K23)	U1.14 (G1)

Step 3 · Logic power (wires 35–48)



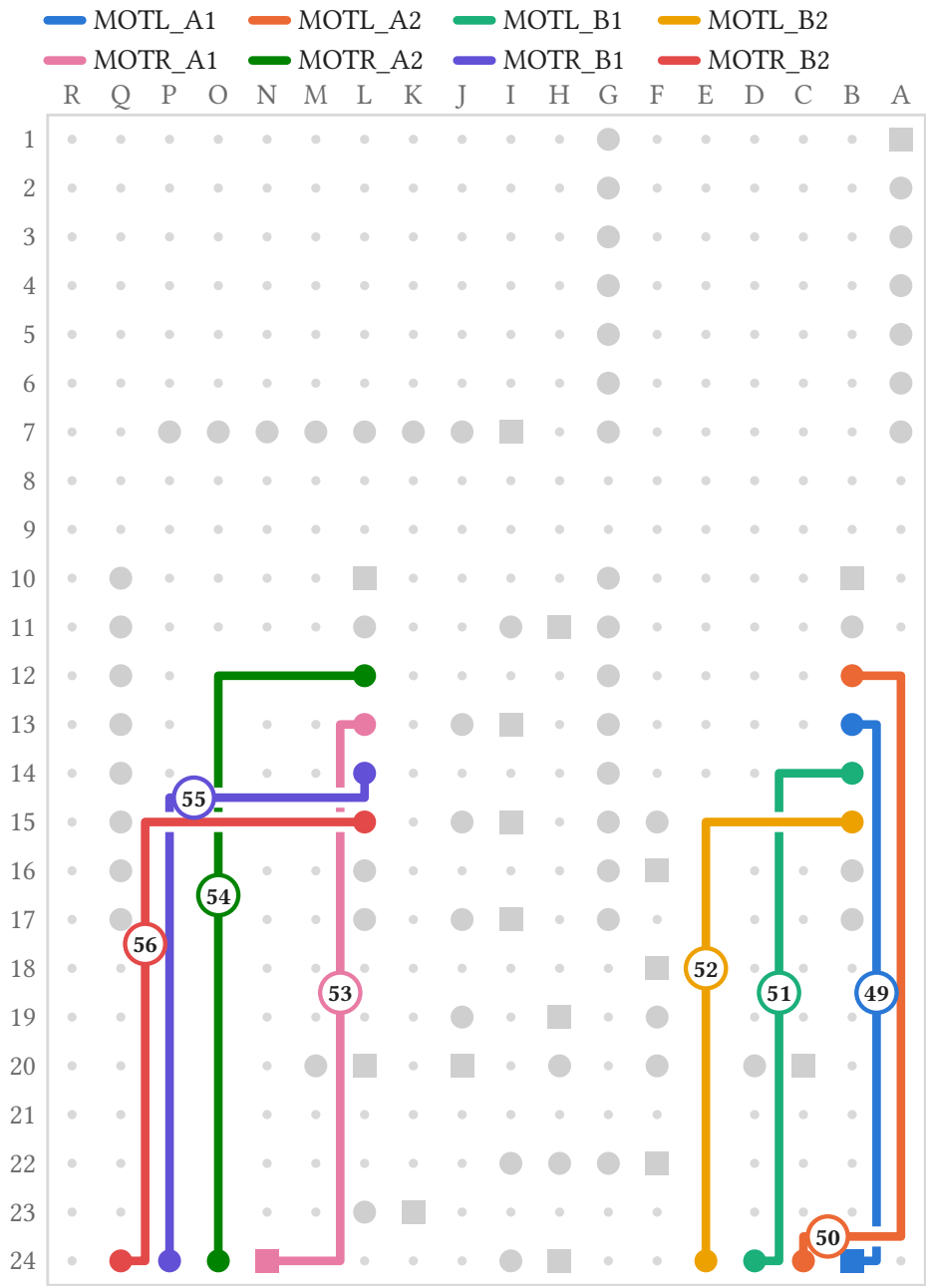
3V3 straps and the shared nENABLE.

	#	Net	Gauge	From	To	Holes
☐	35	+3V3	26–30 AWG	A1.10 (G16)	A1.11 (G15)	1
☐	36	+3V3	26–30 AWG	A1.11 (G15)	A1.13 (G13)	3
☐	37	+3V3	26–30 AWG	A1.13 (G13)	A1.14 (G12)	1
☐	38	+3V3	26–30 AWG	A1.14 (G12)	R1.1 (H11)	2
☐	39	+3V3	26–30 AWG	R1.1 (H11)	splice (H9)	2
☐	40	+3V3	26–30 AWG	splice (H9)	J3.1 (I7)	3
☐	41	+3V3	26–30 AWG	J3.1 (I7)	U1.12 (G3)	6
☐	42	+3V3	26–30 AWG	splice (H9)	A2.14 (Q12)	12
☐	43	+3V3	26–30 AWG	A2.14 (Q12)	A2.13 (Q13)	1
☐	44	+3V3	26–30 AWG	A2.13 (Q13)	A2.11 (Q15)	3
☐	45	+3V3	26–30 AWG	A2.11 (Q15)	A2.10 (Q16)	1
☐	46	NENABLE	26–30 AWG	A1.9 (G17)	R1.2 (I11)	8
☐	47	NENABLE	26–30 AWG	R1.2 (I11)	U1.8 (G7)	6
☐	48	NENABLE	26–30 AWG	A1.9 (G17)	A2.9 (Q17)	11

Continuity after this step

	Net	Probe on	Must beep to
☐	+3V3	A1.10 (G16)	A1.11 (G15), A1.13 (G13), A1.14 (G12), A2.10 (Q16), A2.11 (Q15), A2.13 (Q13), A2.14 (Q12), J3.1 (I7), R1.1 (H11), U1.12 (G3)
☐	NENABLE	A1.9 (G17)	A2.9 (Q17), R1.2 (I11), U1.8 (G7)

Step 4 · Motor coils (wires 49–56)



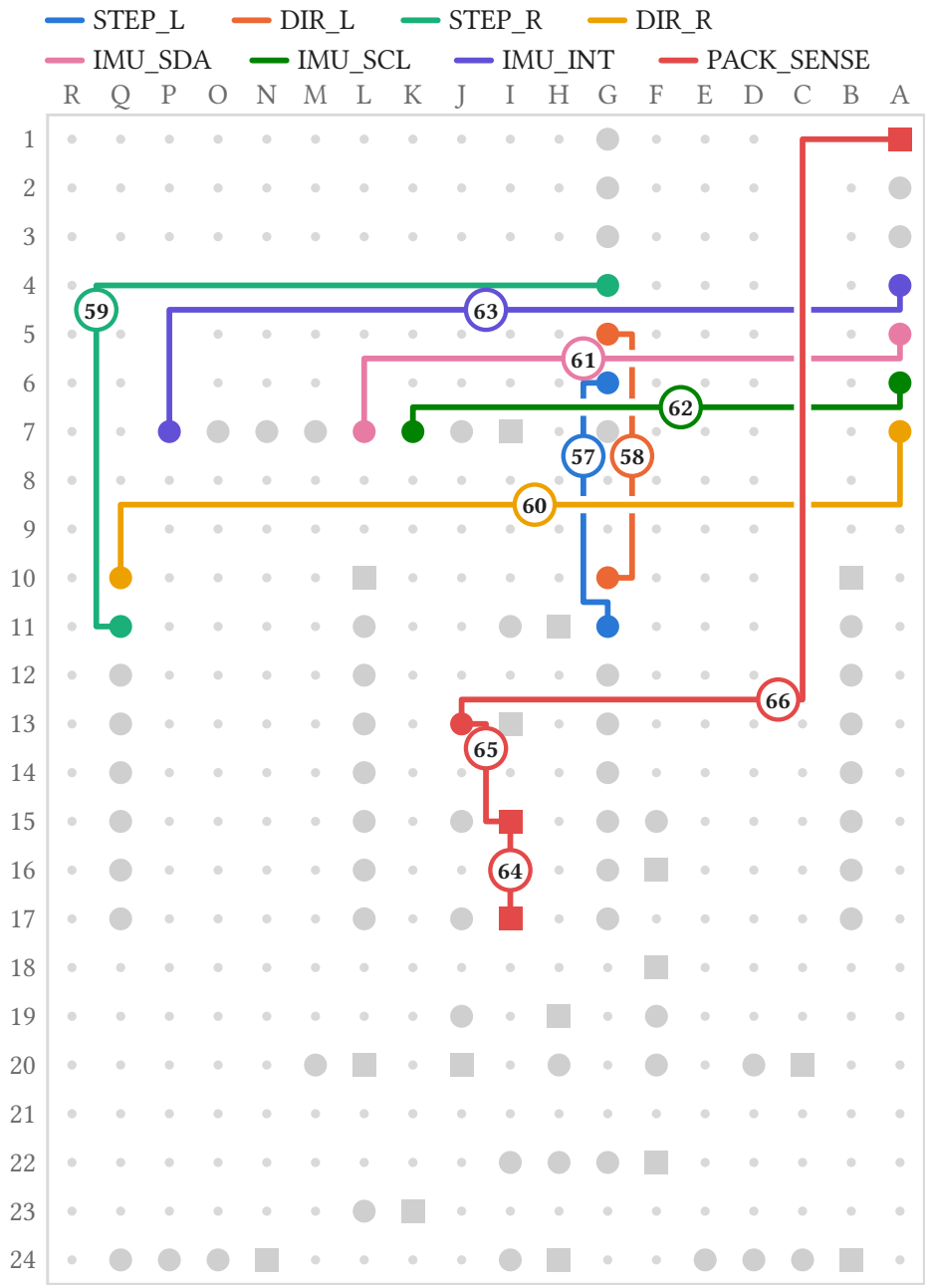
Driver outputs to the motor headers.

	#	Net	Gauge	From	To	Holes
☐	49	MOTL_A1	20 AWG	A1.4 (B13)	J4.1 (B24)	12
☐	50	MOTL_A2	20 AWG	A1.3 (B12)	J4.2 (C24)	15
☐	51	MOTL_B1	20 AWG	A1.5 (B14)	J4.3 (D24)	12
☐	52	MOTL_B2	20 AWG	A1.6 (B15)	J4.4 (E24)	12
☐	53	MOTR_A1	20 AWG	A2.4 (L13)	J5.1 (N24)	13
☐	54	MOTR_A2	20 AWG	A2.3 (L12)	J5.2 (O24)	15
☐	55	MOTR_B1	20 AWG	A2.5 (L14)	J5.3 (P24)	14
☐	56	MOTR_B2	20 AWG	A2.6 (L15)	J5.4 (Q24)	14

Continuity after this step

	Net	Probe on	Must beep to
☐	MOTL_A1	A1.4 (B13)	J4.1 (B24)
☐	MOTL_A2	A1.3 (B12)	J4.2 (C24)
☐	MOTL_B1	A1.5 (B14)	J4.3 (D24)
☐	MOTL_B2	A1.6 (B15)	J4.4 (E24)
☐	MOTR_A1	A2.4 (L13)	J5.1 (N24)
☐	MOTR_A2	A2.3 (L12)	J5.2 (O24)
☐	MOTR_B1	A2.5 (L14)	J5.3 (P24)
☐	MOTR_B2	A2.6 (L15)	J5.4 (Q24)

Step 5 · Signals (wires 57–66)



XIAO to drivers, IMU and the pack divider.

	#	Net	Gauge	From	To	Holes
☐	57	STEP_L	26–30 AWG	A1.15 (G11)	U1.9 (G6)	6
☐	58	DIR_L	26–30 AWG	A1.16 (G10)	U1.10 (G5)	6
☐	59	STEP_R	26–30 AWG	A2.15 (Q11)	U1.11 (G4)	18
☐	60	DIR_R	26–30 AWG	A2.16 (Q10)	U1.7 (A7)	19
☐	61	IMU_SDA	26–30 AWG	J3.4 (L7)	U1.5 (A5)	13
☐	62	IMU_SCL	26–30 AWG	J3.3 (K7)	U1.6 (A6)	11
☐	63	IMU_INT	26–30 AWG	J3.8 (P7)	U1.4 (A4)	18
☐	64	PACK_SENSE	26–30 AWG	C3.1 (I17)	R3.1 (I15)	2
☐	65	PACK_SENSE	26–30 AWG	R3.1 (I15)	R2.2 (J13)	3
☐	66	PACK_SENSE	26–30 AWG	R2.2 (J13)	U1.1 (A1)	21

Continuity after this step

	Net	Probe on	Must beep to
☐	STEP_L	A1.15 (G11)	U1.9 (G6)
☐	DIR_L	A1.16 (G10)	U1.10 (G5)
☐	STEP_R	A2.15 (Q11)	U1.11 (G4)
☐	DIR_R	A2.16 (Q10)	U1.7 (A7)
☐	IMU_SDA	J3.4 (L7)	U1.5 (A5)
☐	IMU_SCL	J3.3 (K7)	U1.6 (A6)
☐	IMU_INT	J3.8 (P7)	U1.4 (A4)
☐	PACK_SENSE	C3.1 (I17)	R2.2 (J13), R3.1 (I15), U1.1 (A1)

Before first power-up

Each pair must not beep. Then read it on the resistance range: a reading that starts low and climbs is a capacitor charging, kilohms is a resistor between the rails, and one that stays near 0 Ω is a short. Fit the modules only when every pair passes.

	Rail	Rail	Probe on	And on
☐	VBAT	XIAO_5V	A1.8 (B17)	D1.1 (K23)
☐	VBAT	+3V3	A1.8 (B17)	A1.10 (G16)
☐	VBAT	GND	A1.8 (B17)	A1.1 (B10)
☐	XIAO_5V	+3V3	D1.1 (K23)	A1.10 (G16)
☐	XIAO_5V	GND	D1.1 (K23)	A1.1 (B10)
☐	+3V3	GND	A1.10 (G16)	A1.1 (B10)